Lecture 27

Bayesian missing data imputation

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(Edited from the lecture notes by Brian J. Reich, NC State)

Recap: Hierarchical models

Often Bayesian models can be written in the following layers of the hierarchy

1. **Data layer**: $[\mathbf{Y}|\theta, \alpha]$ is the likelihood for the observed data **Y** given the model parameters.

2. **Process layer**: $[\theta|\alpha]$ is the model for the parameters $\boldsymbol{\theta}$ that define the latent data generating process.

3. **Prior layer**: $[\alpha]$ prior for hyperparameters.

Missing data models

- We will deal with missing data in the linear regression context, but the ideas apply to all models.
- The model is
$$Y_i \sim \text{Normal}(\beta_0 + \beta_1 X_{i1} + ... + \beta_p X_{ip}, \sigma^2).$$
- Often either Y_i or elements X_{ij} are missing.
- We will study separately the case of missing responses and missing covariates.

Missing responses

- If the response is missing this is essentially a prediction problem.
- We obtain samples from the PPD of Y_i.
- At each MCMC iteration we simply draw

$$Y_i \sim \text{Normal}(\beta_0 + \beta_1 X_{i1} + ... + \beta_p X_{ip}, \sigma^2).$$

- This distribution accounts for random error as well as uncertainty in the model parameters.
- For the other updates the data are essentially complete.

Missing covariates

- Now say all responses are observed, but a some covariates are missing.
- The simplest approach is imputation, e.g., just plug in the sample mean of the covariate for the missing values.
- This doesn't account for uncertainty in the imputations.
- Bayesian methods handle this well using MCMC.

Missing covariates

- The main idea is to treat the missing values as unknown parameters in the Bayesian model.
- Unknown parameters need priors, so missing $\mathbf{X}_i = (X_{i1}, ..., X_{ip})^T$ must have priors such as
$$\mathbf{X}_i \sim \text{Normal}(\boldsymbol{\mu}_X, \boldsymbol{\Sigma}_X).$$
- Assumptions about missing data:
  - Missing status is independent of Y_i and $\mathbf{X}_i$
  - Covariates are Gaussian
- There are ways to relax both assumptions, but it becomes complicated.

Missing covariates

- Of course if the prior is way off, the results will be invalid.
- For example, if in reality the data are not missing at random the Bayesian model will likely give bad results.
- If specified correctly, the model will lead to inference for β that properly accounts for uncertainty about the missing data.

Hierarchical linear regression model with missing data

- $Y_i|\mathbf{X}_i, \boldsymbol{\beta}, \sigma^2 \sim \text{Normal}(\mathbf{X}_i^T\boldsymbol{\beta}, \sigma^2)$
- $\mathbf{X}_i|\boldsymbol{\mu}, \boldsymbol{\Sigma} \sim \text{Normal}_P(\boldsymbol{\mu}, \boldsymbol{\Sigma})$
- $p(\boldsymbol{\beta}) \propto \text{Normal}(0, 100^2\boldsymbol{I}_P)$
- $\sigma^2 \sim \text{Inverse-Gamma}(0.01, 0.01)$
- $\boldsymbol{\mu} \sim \text{Normal}(0, 100^2\boldsymbol{I}_P)$
- $\boldsymbol{\Sigma} \sim \text{Inverse-Wishart}(0.01, 0.01\boldsymbol{I}_P)$

Overview of the Gibbs sampling algorithm

- The full conditional of missing Y_i is:

$$Y_i|\mathbf{X}_i, \boldsymbol{\beta}, \sigma^2 \sim \text{Normal}(\mathbf{X}_i^T\boldsymbol{\beta}, \sigma^2)$$

- The full conditional of missing X_i is:

  conjugate and similar to that of $\boldsymbol{\beta}$, and the derivation follows the same path as that for $\boldsymbol{\beta}$.

- In fact, all the full conditionals are conjugate.

Thank you!